



ParcelPilot Support Agent — Complete Step-by-Step Video Demo Script

10-Minute Video Evaluation • Exact Speech Prompts • Step-by-Step Query Actions • 5 → 0 Live Issues Countdown

Primary Production URL (Render): <https://parcelpilot-agent-05e4.onrender.com>

Secondary Production URL (Railway): <https://parcelpilot-agent-production.up.railway.app>

GitHub Repository: <https://github.com/Sumitboii/-parcelpilot-agent>



Timeline & Agenda Overview

Timecode	Segment	Core Demonstration & Live Issue State
00:00 – 01:15	1. Introduction & Context	Overview of ParcelPilot, support bottlenecks, and live UI layout (5 Live Issues)
01:15 – 02:30	2. Architecture & Multi-Key	Plain Python tool router, Groq multi-key health step function, zero-latency LRU
02:30 – 03:45	3. Foundational Contract Override	Northstar cancellation query (Agreement \$2 overrides Policy & bad TKT-450)
03:45 – 05:00	4. Live Issue 1: TKT-505 P1 SLA	P1 SLA breach triage + interactive confirmation card → Count drops: 5 → 4
05:00 – 06:15	5. Live Issue 2: TKT-501 Northstar SLA	Agreement 15-min SLA breach + escalation confirmation → Count drops: 4 → 3
06:15 – 07:15	6. Live Issue 3: TKT-502 LumenWorks	KI-208 bulk upload limit vs bug workaround → Count drops: 3 → 2
07:15 – 08:15	7. Live Issue 4: TKT-504 SwiftShip	SwiftShip 20-min webhook delay explanation → Count drops: 2 → 1
08:15 – 09:15	8. Live Issue 5: Account Cluster	Cross-ticket correlation between TKT-501 & TKT-504 → Count drops: 1 → 0 (All Resolved)
09:15 – 10:00	9. Trust, Metrics & Wrap-Up	Grounded citations, 66/66 test suite, hallucination prevention



Detailed Step-by-Step Script: Words to Speak & Screen Actions

Segment 1: Introduction & Executive Context

00:00 – 01:15



STEP-BY-STEP ON-SCREEN ACTIONS:

- Open browser tab to the live deployment: <https://parcelpilot-agent-05e4.onrender.com> .
- Move cursor to the top bar: point out the active role indicator (**Rohit — Support Agent**) and the green **Connected** status badge.
- Move cursor to the left sidebar: highlight the **Live Issues** component showing **5** active items in red/amber badges.
- Hover over the central conversation area showing the empty state suggestion chips.



EXACT WORDS TO SPEAK:

“Hello everyone! Today I’m excited to present the **ParcelPilot AI Support Copilot** — an enterprise-grade operational assistant built for ParcelPilot’s support and customer success teams.

ParcelPilot coordinates freight and parcel deliveries across multiple logistics carriers, processing hundreds of complex queries daily. Today, support agents lose hours manually cross-referencing multi-page PDFs, custom enterprise contracts, order spreadsheets, and legacy ticketing records.

The hardest part of this domain isn’t just fetching data — it’s handling **authority conflicts, speed, and trust**. Customer agreements override standard policies, old policies get deprecated, rate limits cause unexpected service outages, and historical ticket resolutions often contain errors.

We built this production system to solve those exact operational headaches: delivering sub-second, trustworthy, cited answers with automated multi-key failover while keeping human agents strictly in control of critical business actions.

Notice on the top-left sidebar: our proactive radar has automatically surfaced **5 Live Issues** that need our attention. Let’s look at how the system works under the hood and resolve every single one of them.”

Segment 2: Architecture & Multi-Key Health Step Function

01:15 – 02:30

STEP-BY-STEP ON-SCREEN ACTIONS:

1. Scroll slightly or open developer tools / docs tab showing API endpoints (`/health` , `/chat` , `/proactive`).
2. Point out the sub-second streaming response capability and zero-framework lean architecture.
3. Move back to the main chat window ready to trigger queries.

EXACT WORDS TO SPEAK:

"Let's take a quick look at the architecture powering ParcelPilot.

First, **Multi-Key Health Step Function**: To guarantee zero downtime and eliminate rate-limit disruptions, we built a custom `GroqKeyPoolManager` . It automatically discovers all available Groq API keys, dispatches requests with zero added latency using an in-memory Least-Recently-Used rotation, and includes an automated step function that fails over to a backup key in milliseconds if a 429 rate limit occurs. A background worker continuously probes key health every 30 seconds.

Second, **Locked Source-Authority Hierarchy**: The agent enforces a deterministic 4-tier precedence hierarchy:

1. Level 1: **Signed Customer Agreements** always take top priority and override general rules for that specific account.
2. Level 2: **Current Support Policies** (`01_Support_Policy_v3_CURRENT.pdf`) come second.
3. Level 3: **Standard Operating Procedures (SOPs)** and operations guides come third.
4. Level 4: **Historical Ticket Resolutions** are treated strictly as unverified context — any known historical error is actively rejected.
5. Any deprecated documents — such as `02_Support_Policy_v2_DEPRECATED.pdf` — are completely filtered out.

Third, **Safety & Privacy**: Every data query passes through an in-memory data store with Role-Based Access Control, and state-changing actions like escalations are intercepted by an interactive confirmation gate."

Segment 3: Foundational Demo — Source Hierarchy & Contract Override

02:30 – 03:45

STEP-BY-STEP ON-SCREEN ACTIONS:

1. In the central chat area, click the suggestion chip: **Northstar cancellation** (or type: *'Can Northstar cancel ORD-1001 without a fee? Explain why.'*).
2. Press **Enter** or click **Send**.
3. Point out the workflow card appearing with animated tool chips: `data_lookup` and `document_search` .
4. Highlight the formatted Markdown answer, the cited contract section, and the explicit rejection of legacy ticket `TKT-450` .

EXACT WORDS TO SPEAK:

"Let's begin with a foundational contract conflict scenario.

I'll ask: *'Can Northstar cancel ORD-1001 without a fee? Explain why.'*

Watch the execution flow:

1. The agent calls `data_lookup` to check `ORD-1001` . It verifies that the order is still in `BOOKED` status and has not been dispatched.
2. It queries `document_search` for Northstar's specific enterprise agreement.

Look at the answer:

- The agent confirms: **Yes, Northstar can cancel with zero fee.**
- It cites **Section 2 of Northstar's Enterprise Agreement**, which explicitly waives all cancellation fees for shipments cancelled before pickup.
- Most importantly, it actively points out that a past ticket resolution — **TKT-450** — incorrectly charged the customer ₹250. The agent correctly rejected that historical mistake in favor of the authoritative signed contract.
- Every single claim is backed by a clickable citation badge."

Segment 4: Live Issue 1 — TKT-505 Axis Labs P1 SLA Breach

03:45 – 05:00

Live Issues: 5 → 4

STEP-BY-STEP ON-SCREEN ACTIONS:

1. Direct attention to the left sidebar showing **5** in the issue counter.
2. Click the top sidebar item: **TKT-505 — Axis Labs** (or click chip **TKT-505 API key**).
3. Watch the agent stream the SLA calculation, classify as **P1**, and generate the interactive ⚡ **Confirm Action** card.
4. Click the blue **Confirm** button on the confirmation card.
5. Point out the green confirmation message: ✓ **Escalation created: ESC-20260816-XXXX**.
6. **Point to the top-left sidebar:** Show that **TKT-505 is now marked with ✓ RESOLVED** and the **counter decremented from 5 to 4**.

EXACT WORDS TO SPEAK:

"Now let's tackle our first live issue on the left sidebar: **TKT-505 for Axis Labs**.

I'll click **TKT-505** directly on the sidebar.

Notice how the agent analyzes this incident:

1. It checks the ticket timestamps: Opened at 08:30 IST, snapshot is 11:00 IST — **2 hours and 30 minutes have elapsed**.
2. It checks governing policy: Axis Labs is an Enterprise account with no custom agreement, so Standard Policy v3 applies. For suspected API key / credential exposure, the classification is **P1**, with a strict **30-minute response target**.
3. The SLA is **breached by 2 hours**. Policy mandates immediate escalation and key revocation.

But instead of silently modifying the database, look at the screen: The agent rendered an interactive **Confirmation Card** with structured fields: Ticket ID, Severity, Reason, and Assignee.

I will now click **Confirm**.

The escalation is written to our audit log with ID **ESC-20260816-0001**.

And look at the top-left sidebar: **TKT-505 is immediately marked ✓ RESOLVED with a green badge, and our Live Issues counter drops from 5 to 4!**"

Segment 5: Live Issue 2 — TKT-501 Northstar P1 SLA Breach

05:00 – 06:15

Live Issues: 4 → 3

STEP-BY-STEP ON-SCREEN ACTIONS:

1. In the left sidebar, click the next item: **TKT-501 — Northstar Logistics**.
2. Watch the agent retrieve Northstar's agreement, detect the 15-min P1 target, and calculate the 15-minute breach.
3. Observe the rendered ⚡ **Confirm Action** escalation card assigned to *Lead Integrations Engineer*.
4. Click the **Confirm** button.
5. **Point to the top-left sidebar:** Show that **TKT-501 is marked with ✓ RESOLVED** and the **counter decremented from 4 to 3**.

EXACT WORDS TO SPEAK:

"Let's resolve our second live issue: **TKT-501 for Northstar Logistics**.

I'll click **TKT-501** on the sidebar.

Here we see another critical contract override:

1. Standard Enterprise policy allows 30 minutes for P1 incidents.
2. But Northstar's signed agreement **Section 4** guarantees an expedited **15-minute response target**.
3. The ticket was opened at 10:30 IST (30 minutes elapsed), meaning the SLA is **breached by 15 minutes**.

The agent immediately generates an escalation card targeted to the Lead Integrations Engineer.

I click **Confirm**.

The escalation is logged, and notice our sidebar: **TKT-501 is now marked ✓ RESOLVED , and the counter decrements from 4 to 3!**"

Segment 6: Live Issue 3 — TKT-502 LumenWorks KI-208 Bug Workaround

06:15 – 07:15

Live Issues: 3 → 2

STEP-BY-STEP ON-SCREEN ACTIONS:

1. In the left sidebar, click **TKT-502 — LumenWorks** (or click chip `Bulk upload limit`).
2. Watch the agent run `data_lookup` on ticket **TKT-502** and `document_search` on the Product Operations Guide.
3. Highlight the distinction between the **5,000-row product limit** and the **KI-208 bug workaround (split below 3,000 rows)**.
4. **Point to the top-left sidebar:** Show that **TKT-502 is marked with ✓ RESOLVED** and the **counter decremented from 3 to 2**.

EXACT WORDS TO SPEAK:

"Now let's look at our third issue: **TKT-502 for LumenWorks**, reporting a bulk upload failure on a 4,200-row CSV.

I'll click **TKT-502** on the sidebar.

Look at how intelligently the agent separates product specifications from temporary bugs:

1. It cites the official Product Operations Guide: the true product capacity is **5,000 rows per CSV**.
2. But it identifies active bug **KI-208**, which causes intermittent 504 gateway timeouts on files above ~3,000 rows.
3. It gives the exact customer communication: advise the customer to split their 4,200-row file into two batches of 2,100 rows while engineering patches the parser.
4. It also rejects legacy ticket **TKT-451**, which erroneously told the customer 3,000 was their plan ceiling.

With that resolved, look at the sidebar: **TKT-502 turns green with ✓ RESOLVED**, and the **counter drops from 3 to 2!**"

Segment 7: Live Issue 4 — TKT-504 SwiftShip Webhook Lag (KI-211)

07:15 – 08:15

Live Issues: 2 → 1

STEP-BY-STEP ON-SCREEN ACTIONS:

1. In the left sidebar, click **TKT-504 — Northstar Logistics** (SwiftShip pickup lag).
2. Watch the agent evaluate order `ORD-1001` and retrieve Known Issue **KI-211**.
3. Highlight the explicit guardrail: **Do not conclude pickup failed**.
4. **Point to the top-left sidebar:** Show that **TKT-504 is marked with ✓ RESOLVED** and the **counter decremented from 2 to 1**.

EXACT WORDS TO SPEAK:

"Our fourth live issue is **TKT-504**: a ticket where the customer says the SwiftShip driver collected the parcel 10 minutes ago, but the portal still shows `BOOKED` .

I'll click **TKT-504** on the sidebar.

Watch the agent's reasoning:

- A naive agent might assume the pickup failed and trigger an unnecessary redelivery.
- But our agent identifies **Known Issue KI-211**: SwiftShip pickup confirmation webhooks experience an upstream carrier delay of up to **20 minutes**.
- It instructs the support agent: **Do not tell the customer pickup failed**. Advise them to wait for the 20-minute webhook window to expire or check carrier tracking directly.

As the response completes: **TKT-504 is marked ✓ RESOLVED**, and the **Live Issues counter drops from 2 to 1!**"

Segment 8: Live Issue 5 — Account Cluster Pattern Analysis

08:15 – 09:15

Live Issues: 1 → 0 (All Done!)

STEP-BY-STEP ON-SCREEN ACTIONS:

1. In the left sidebar, click the final item: **Account Cluster — Northstar Logistics**.
2. Watch the agent synthesize open tickets across TKT-501 and TKT-504 into an integrated account pattern table.
3. **Point to the top-left sidebar:** Show that **Account Cluster is marked with ✓ RESOLVED**, the counter badge displays **0** in bright green, and the success toast appears: 🎉 All 5 Live Issues Resolved! .

EXACT WORDS TO SPEAK:

"Now let's resolve our final live issue: **Account Cluster for Northstar Logistics**.

I'll click **Account Cluster** on the sidebar.

Operations leads need cross-ticket visibility. The agent analyzes Northstar's account history:

- It correlates **TKT-501** (webhook verification failure) and **TKT-504** (tracking status lag).
- It identifies the root cause: an integration sync degradation between Northstar's dispatch automation and SwiftShip carrier webhooks.
- It recommends assigning both tickets to the same technical specialist to prevent duplicate troubleshooting.

And look at the top-left sidebar:

The counter reaches 0, the badge turns solid green, and all 5 Live Issues are completely resolved!"

Segment 9: Trust, Evaluation Metrics & Wrap-Up

09:15 – 10:00

STEP-BY-STEP ON-SCREEN ACTIONS:

1. Scroll through the conversation showing clean Markdown tables, green confirmation cards, and citation badges.
2. In the top right, switch the role dropdown to **Priya Mehta — CSM / Escalation Manager** to show role-based access control.
3. Show the automated test suite benchmark passing **66/66 tests**.

EXACT WORDS TO SPEAK:

"To summarize what we've seen:

- **100% Grounded & Cited:** Every single answer cites the authoritative signed agreement, policy PDF, or database record. The agent never hallucinates.
- **Speed & High Availability:** Powered by our multi-key Groq pool with zero-latency failover and a lightweight in-memory vector store that runs comfortably in under 40 megabytes of RAM.
- **Human Control:** High-impact state changes are protected behind interactive confirmation cards.
- **Proactive Operations:** Automated SLA breach detection and known issue grouping with live tracking from 5 down to 0.
- **Verified Quality:** 66 out of 66 automated tests pass across unit tests, key rotation, and full evaluation matrices.

Thank you for your time, and I look forward to your questions!"